COMPOSITIONAL DIFFERENCE-IN-DIFFERENCES

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Abstract. Many policy evaluations involve vectors of category-specific quantities, either through aggregation of categorical outcomes (e.g., employment type, major choice) or compositional measures (e.g., GDP by sector, votes by party, electricity generation by source). In such settings, beyond the quantities, treatment effects on both the within-category shares and the overall total across categories are often economically relevant. Standard approaches treat each category as a separate binary outcome and apply DiD independently. This often yields unrealistic counterfactual shares, fails to recover compositional effects and effects on totals, and lacks a clear economic interpretation. I develop Compositional Difference-in-Differences (CoDiD), a framework that jointly identifies treatment effects on totals and shares in an economically coherent way. The key assumption is parallel growth: absent treatment, vectors of log-quantities would have evolved in parallel across groups. Under a random-utility discrete-choice model, this is equivalent to parallel trends in expected utilities, so the average change in latent attractiveness is the same for treated and control units. Geometrically, counterfactual shares evolve along parallel paths in the probability simplex. For multiple periods, I also provide a relaxation that yields informative bounds. I illustrate the approach using the impact of early voting reforms on the 2008 U.S. election.

Keywords: categorical outcomes, difference-in-differences, causal inference, compositional data analysis, probability simplex, treatment effects.

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Introduction

Many causal questions in Difference-in-Differences (DiD) settings involve vectors of categorical quantities that together form a well-defined total. These arise naturally when individual-level categorical outcomes are aggregated. For instance, in evaluating a minimum wage policy, individuals may be classified as unemployed, part-time employed, or full-time employed, and the compositional vector counts how many fall into each group. In other cases, the data consist directly of quantities across categories, with no underlying individual identifiers, such as the number of votes received by each political party in an election. Compositional vectors also appear when a single quantity is decomposed into meaningful parts, as in electricity generation by source (gas, coal, nuclear, renewable), GDP by sector (agriculture, industry, services), or a budget allocated to each category (for example, food, rent, healthcare, others). In all such cases, the outcome is multidimensional, and beyond the quantity both the shares and the total across categories can matter for the analysis.

The canonical empirical strategy is to treat each category separately, either as a binary outcome at the individual level or as a continuous quantity in compositional data, and estimate standard DiD regressions. This approach ignores the joint structure of the outcome and analyzes each category in isolation. I argue that it suffers from several compounding deficiencies. When the object of interest is the effect on shares, linear DiD imposes a parallel trends assumption in levels, implying that shares would change by the same absolute amount across groups. This is inconsistent with the simplex constraint, since shares are bounded between zero and one, and can lead to infeasible counterfactuals, especially when baseline shares are high. This issue is well acknowledged (Lechner, 2011; Athey and Imbens, 2006; Wooldridge, 2023; Ahn and Kasahara, 2025). Moreover, estimating separate DiD models for each category generally produces counterfactual shares that do not sum to one, forcing arbitrary normalization choices. As a result, this approach cannot capture how treatment reallocates probability mass across mutually exclusive categories. Alternative methods, such as geodesic DiD (Zhou et al., 2025), transition-independence approaches (Ahn and Kasahara, 2025), or nonlinear DiD (Wooldridge, 2023), avoid infeasible counterfactuals but have other important limitations.

In many applications, treatment effects across categories and on totals are jointly relevant. A voting reform may reshape vote shares while changing turnout; a minimum wage may

alter employment composition and the total number of workers; an education reform may shift students across majors while affecting the number of graduates; and a carbon policy may change both the energy mix and total electricity generation. Existing methods require estimating these effects separately, often under different models and assumptions, and cannot jointly and consistently recover effects on shares and totals. Yet many empirical questions demand a framework that coherently captures both absolute and relative changes. Moreover, alternatives to linear DiD often lack clear economic foundations: their counterfactuals for categorical outcomes are not generally consistent with behavioral or structural models, such as utility maximization or probabilistic choice, making results harder to interpret and less credible in applied work.

I develop *Compositional Difference-in-Differences* (CoDiD), a framework that solves all the issues mentioned simultaneously. The variable I consider is an aggregated compositional vector recording the count or quantity in each of p categories. CoDiD identifies the *entire* counterfactual vector, and from it derives treatment effects on both the intensive margin (shares) and the extensive margin (total). The key identifying condition is *parallel growth*: in the absence of treatment, the proportional growth rate of each category would be identical in the treated and control groups. Formally, the vector of log-quantities satisfies a standard parallel trends condition. This assumption has three complementary interpretations that I develop in full.

First, under a standard multinomial random-utility model (RUM), parallel growth is equivalent to parallel trends in latent expected utilities. Absent treatment, both groups experience the same evolution in the average attractiveness of each option. The treatment effect measures the policy-induced shift in the underlying desirability of each category. Second, parallel growth implies parallel trends in the log-odds (multinomial logit) transform of shares. The counterfactual share vector is recovered by the softmax of the DiD formula applied to log-odds. This guarantees that counterfactual shares are always non-negative, always sum to one, and respect the simplex geometry by construction. Finally, from a geometric perspective, each share vector corresponds to a point in the simplex, and its evolution traces a path within this constrained space. The parallel growth assumption implies that, absent treatment, the treated group's counterfactual path would be parallel to the control

group's path under Aitchison geometry, the standard geometric framework for compositional data (Aitchison (1982); Egozcue et al. (2003)).

The counterfactual is identified by a closed-form formula and delivers point identification of all treatment effect parameters of interest. The plug-in estimator substitutes sample counts into the identification formula. Under multinomial sampling with cell sizes growing proportionally, I prove that all CoDiD estimators are $\sqrt{n}$-consistent and asymptotically normal. I derive exact analytic expressions for the asymptotic variance using the multivariate delta method, and I prove that the parametric bootstrap consistently estimates the limiting distribution. I also propose a formal Wald pre-trends test. When multiple pre-treatment periods are available and the pre-trends test raises concerns, I derive sharp bounds on the counterfactual vector by requiring the post-treatment log-difference to lie within the convex hull of observed pre-treatment log-differences. These bounds remain informative and preserve the geometric structure.

I apply CoDiD to evaluate early-voting reform on the 2008 U.S. presidential election, comparing Maryland and New Jersey (treated) with Pennsylvania and New York (control). The treatment increased total turnout by 4.4%, raised the Democratic share by 0.92 percentage points, and reduced the Republican share by 1.17 percentage points.

Related Literature. This paper contributes to several strands of literature.

Difference-in-Differences. The classical DiD literature focuses on scalar outcomes (Lechner, 2011; Cafri et al., 2019; De Chaisemartin and d'Haultfoeuille, 2023; Roth et al., 2023; Baker et al., 2025). Recent methodological advances include staggered treatment adoption (Callaway and Sant'Anna, 2020), synthetic control hybrids (Arkhangelsky et al., 2021), and relaxations of parallel trends (Manski and Pepper, 2018; Rambachan and Roth, 2020; Ban and Kédagni, 2022). A growing literature extends DiD to distributional outcomes: quantile DiD (Meyer et al., 1990), changes-in-changes (Athey and Imbens, 2006), multivariate outcomes (Torous et al., 2021), CDF-based methods (Havnes and Mogstad, 2015; Roth and Sant'Anna, 2021), copula approaches (Callaway et al., 2018; Callaway and Li, 2019; Ghanem et al., 2023), and characteristic-function methods (Bonhomme and Sauder, 2011). None of these frameworks addresses the specific geometry of compositional outcomes constrained to the simplex.

The closest predecessor for categorical DiD is Graves et al. (2022), who imposes a linear parallel trends assumption directly on proportions. As I show above, this is generally infeasible. Zhou et al. (2025) extends DiD to non-Euclidean data using Fréchet means and geodesic transport; the approach is mathematically elegant but does not jointly recover effects on totals and lacks an economic foundation in terms of choice primitives. Tchetgen Tchetgen et al. (2024) consider a DiD framework for count data in the binomial setting; CoDiD extends this to the multinomial case with $p \geq 2$ unordered categories. Wooldridge (2023) proposes fractional logit DiD for a single proportion; my framework handles the full compositional vector jointly.

Compositional Data Analysis. CoDA was pioneered by Aitchison (1982, 1990, 1992, 2002) and further developed by Egozcue et al. (2003); Billheimer et al. (2001); Barceló-Vidal et al. (2001). Arnold et al. (2020) connect CoDA to causal inference in a general framework. CoDiD bridges CoDA with the econometric DiD identification strategy, providing a treatment-effects interpretation of compositional change.

Random Utility Models. The economic foundation of parallel growth rests on the standard multinomial RUM (McFadden, 1972, 1974, 1977) and its GEV generalizations. I show that the parallel growth assumption has a precise structural interpretation as parallel trends in expected utilities, connecting CoDiD to the broader literature on discrete choice and structural microeconometrics.

Paper Organization. Section 1 develops the canonical 2×2 case, introducing the framework, the parallel growth assumption, and its three interpretations. Section 2 extends the framework to multiple periods and introduces the partial-identification relaxation. Section 3 provides full asymptotic theory, bootstrap consistency, and a pre-trends test. Section 4 presents the empirical application. Section 5 concludes. Appendices collect proofs, covariate adjustment, computational steps, and GEV extensions.

1. Canonical 2×2 DiD Case

1.1. **Framework and Notation.**

Data structure. For each group $g \in \{0,1\}$ and time period $t \in \{0,1\}$, I observe a p-vector of non-negative integer counts (or non-negative quantities):

$$q_{g,t} = \big(q_{1,g,t},\, q_{2,g,t},\, \ldots,\, q_{p,g,t}\big) \in \mathbb{R}^p_+.$$

Group $g = 1$ is the treated group; $g = 0$ is the control. Period $t = 0$ is pre-treatment; $t = 1$ is post-treatment. There are $p \geq 2$ mutually exclusive and exhaustive categories, so the total count is $n_{g,t} = \sum_{k=1}^{p} q_{k,g,t}$ and the share vector is $\pi_{g,t} = q_{g,t}/n_{g,t} \in \mathcal{S}^{p-1}$, where $\mathcal{S}^{p-1} = \{\pi \in \mathbb{R}^p_+ : \sum_k \pi_k = 1\}$ is the $(p-1)$-dimensional probability simplex.

Potential outcomes. Following the potential outcomes framework, for each category k, group g, and period t:

- $q^0_{k,g,t}$ denotes the *untreated* potential quantity (the quantity that would obtain in category k if group g were untreated at time t),
- $q^1_{k,g,t}$ denotes the *treated* potential quantity.

The corresponding untreated total and share are $n^0_{g,t} = \sum_k q^0_{k,g,t}$ and $\pi^0_{k,g,t} = q^0_{k,g,t}/n^0_{g,t}$. From the data, I can identify the vectors $q^0_{0,0}$ for the control group and $q^0_{1,0}$ for the treated group in the pre-treatment period. In the post-treatment period, I can identify from data the untreated vector of quantities for the control group, $q^0_{0,1}$, and the treated counterpart for the treated group, $q^1_{1,1}$. The counterfactual of interest is the vector of quantities that I would have observed for the treated group in the absence of treatment $q^0_{1,1}$. Identifying this missing object is the core challenge of my analysis. Once this is identified, I can consider the treatment effect parameters.

1.2. **Treatment Effect Parameters.** Because the outcome is a vector rather than a scalar, several distinct and complementary treatment effects can be defined. I introduce them here in order of increasing complexity.

Growth Treatment Effect on the Treated (GTT).. The GTT measures the causal proportional change in the absolute size of each category:

$$\mathrm{GTT}_k = \frac{q^1_{k,1,1}}{q^0_{k,1,1}} - 1 \quad (k = 1, \ldots, p), \qquad \mathrm{GTT} = \frac{n^1_{1,1}}{n^0_{1,1}} - 1. \tag{1}$$

$\mathrm{GTT}_k = 0$ means the treatment had no effect on the absolute count in category k; $\mathrm{GTT}_k > 0$ means the treatment caused that category to grow. The aggregate GTT captures the proportional change in the total due to treatment.

Average Treatment Effect on the Treated (ATT).. The ATT measures the causal change in the *share* allocated to each category:

$$\mathrm{ATT}_k = \pi^1_{k,1,1} - \pi^0_{k,1,1} \quad (k = 1, \ldots, p), \qquad \mathrm{ATT} = \pi^1_{1,1} - \pi^0_{1,1} \in \mathbb{R}^p. \tag{2}$$

Since $\sum_k \pi^1_{k,1,1} = \sum_k \pi^0_{k,1,1} = 1$, the ATT automatically satisfies $\sum_k \mathrm{ATT}_k = 0$: what one category gains, the others collectively lose. This is the absolute redistribution of probability mass across categories.

Compositional Treatment Effect on the Treated (CTT).. The CTT uses the compositional difference operator from compositional data analysis to capture the *relative* redistribution of probability mass. For each category k, let $r_k = \pi^1_{k,1,1}/\pi^0_{k,1,1}$ be the ratio of observed to counterfactual share. Normalizing these ratios to sum to one defines the CTT:

$$\mathrm{CTT} = \pi^1_{1,1} \ominus \pi^0_{1,1} = \frac{1}{\sum_{j=1}^p r_j}\big(r_1, \ldots, r_p\big). \tag{3}$$

A component $\mathrm{CTT}_k > 1/p$ indicates that category k gained relative importance due to treatment; a component $\mathrm{CTT}_k < 1/p$ indicates a relative loss. When there is no compositional treatment effect (all ratios equal one), $\mathrm{CTT} = (1/p, \ldots, 1/p)$, the neutral element of the Aitchison composition. The CTT is invariant to the overall scale of the effect, isolating pure redistribution from size effects.

Functional Treatment Effect on the Treated (FTT).. For any measurable functional $H\colon \mathcal{S}^{p-1} \to \mathbb{R}$, the FTT is defined as:

$$\mathrm{FTT}(H) = H(\pi^1_{1,1}) - H(\pi^0_{1,1}). \tag{4}$$

A leading example is the Herfindahl–Hirschman Index (HHI), $H(\pi) = \sum_k \pi_k^2$, which measures market concentration. The correct procedure is to recover $\pi^0_{1,1}$ via CoDiD and then compute FTT(HHI). Applying DiD directly to $\mathrm{HHI}(\pi)$ is incorrect because it ignores the simplex constraint and conflates the concentration effect with compositional reallocation. For differentiable H, the delta method provides straightforward inference for the FTT; see Section **??**.

Remark 1 (Relationship between GTT and ATT)**.** *The GTT and ATT are complementary but distinct. The GTT measures absolute category-level changes and captures the extensive margin effect. The ATT measures relative reallocation and captures the intensive margin effect. Neither implies the other: a treatment can increase the total without changing any*

share (uniform GTT with zero ATT), or it can completely reshuffle shares without affecting the total (zero aggregate GTT with non-zero ATT). CoDiD recovers both from the same identified counterfactual $q^0_{1,1}$.

1.3. **Identification: Parallel Growth.**

Assumption 1 (Positivity)**.** *For all* $g \in \{0,1\}$ *and* $t \in \{0,1\}$, $q^0_{k,g,t} > 0$ *for every* $k = 1, \ldots, p$.

Assumption 1 requires that every category has positive mass in every group and every period in the absence of treatment. It ensures that log-quantities are well defined and guarantees that the counterfactual formula produces a strictly positive vector. See Remark 2 for the case of zero counts.

Remark 2 (Zero counts)**.** *If* $q_{k,g,t} = 0$ *for some* (k, g, t)*, one may either (i) exclude category* k *from the analysis and restrict to the common support, or (ii) apply a small additive smoothing (e.g., add* $1/2$ *to each count prior to taking logarithms), following standard practice for log-linear models.*

Assumption 2 (No Anticipation)**.** *The treatment has no effect on the treated group in the pre-treatment period:* $q^1_{k,1,0} = q^0_{k,0,0} \equiv q_{k,1,0}$ *for all* k.

Assumption 2 is standard in DiD (Callaway and Sant'Anna, 2020). It implies that the treated group's pre-treatment counts $q_{1,0}$ are observed and equal to their untreated potential outcome, so the identification formula can use observed pre-treatment data for the treated group without contamination from anticipatory behavior.

Because category sizes can differ dramatically in scale, imposing linear parallel trends on raw quantities which would require the same absolute change in large and small categories alike is generally implausible. I instead impose parallel trends on the log of the quantities, which is a proportional (multiplicative) rather than additive restriction.

Assumption 3 (Parallel Growth)**.** *In the absence of treatment, the vector of log-quantities for the treated and control groups evolves in parallel:*

$$\log q^0_{k,1,1} - \log q^0_{k,1,0} = \log q^0_{k,0,1} - \log q^0_{k,0,0} \quad \textit{for all } k = 1, \ldots, p. \tag{5}$$

Assumption 3 states that, absent treatment, the proportional growth rate of each category k would be the same in the treated and control groups. It is the multiplicative counterpart of the additive parallel trends assumption standard in DiD. Like its additive analogue, parallel growth is not testable from the post-treatment data alone, but in settings with multiple pre-treatment periods its pre-treatment implications are testable (Section 3.3).

Figure 1 illustrates the identification logic. The control group's time trend (blue arrow) is extrapolated to the treated group's pre-treatment level to construct the counterfactual (dashed blue arrow). The treatment effect (orange double arrow) is the gap between the observed treated outcome $q^1_{1,1}$ and this counterfactual $q^0_{1,1}$. This reflects the intuitive idea

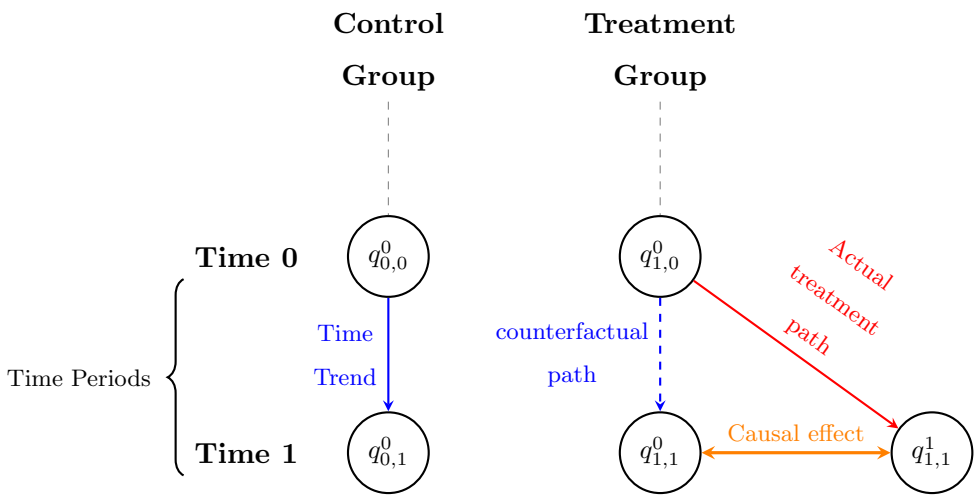

Figure 1. Visual representation of the DiD identification strategy. The counterfactual outcome for the treated group in the post-treatment period ($q^0_{1,1}$) is constructed by extrapolating the time trend observed in the control group (blue arrows) to the treated group's pre-treatment level. The causal effect is the difference between the observed treated outcome ($q^1_{1,1}$) and this counterfactual ($q^0_{1,1}$), shown by the orange double-headed arrow.

that underlying trends affect all categories proportionally to their size. This may appear as a more realistic and flexible assumption than one based on raw quantities in many economic contexts, especially when the sizes of categories differ significantly.

Theorem 1 (Point Identification)**.** *Under Assumptions 2–3, the counterfactual compositional vector and its derived objects are identified:*

$$q^0_{k,1,1} =\equiv \exp\big(\log q_{k,1,0} + \log q_{k,0,1} - \log q_{k,0,0}\big), \tag{6}$$

$$n^0_{1,1} = \sum_{k=1}^{p} \exp\big(\log q_{k,1,0} + \log q_{k,0,1} - \log q_{k,0,0}\big), \tag{7}$$

$$\pi^0_{k,1,1} = \frac{\exp\big(\log q_{k,1,0} + \log q_{k,0,1} - \log q_{k,0,0}\big)}{\sum_{k=1}^{p} \exp\big(\log q_{k,1,0} + \log q_{k,0,1} - \log q_{k,0,0}\big)}. \tag{8}$$

This identification result is constructive and provides a straightforward way to implement the method. Before I present the economic and geometric implication of parallel growths, I show how this assumption restricts the evolution of each margin.

1.4. **Implications for Shares, Totals, and the Aggregation Correction.** The parallel growth assumption simultaneously restricts the evolution of both margins.

Restriction on shares. Define the log-odds (multinomial logit) transformation $\ell\colon \mathcal{S}^{p-1} \to \mathbb{R}^{p-1}$ with baseline category p:

$$\ell(\pi) = \Big(\log \tfrac{\pi_1}{\pi_p},\ \ldots,\ \log \tfrac{\pi_{p-1}}{\pi_p}\Big), \tag{9}$$

with right inverse (softmax):

$$\ell^{-1}(y) = \frac{1}{1 + \sum_{i=1}^{p-1} e^{y_i}} \big(e^{y_1}, \ldots, e^{y_{p-1}}, 1\big). \tag{10}$$

Proposition 1 (Parallel trends in log-odds)**.** *Under Assumptions 1 and 3:*

$$\ell(\pi^0_{1,1}) - \ell(\pi^0_{1,0}) = \ell(\pi^0_{0,1}) - \ell(\pi^0_{0,0}). \tag{11}$$

Equivalently, the counterfactual share vector is:

$$\pi^0_{1,1} = \ell^{-1}\big(\ell(\pi_{1,0}) + \ell(\pi_{0,1}) - \ell(\pi_{0,0})\big). \tag{12}$$

The counterfactual shares in (12) are always in the interior of the simplex by construction, because softmax maps any $\mathbb{R}^{p-1}$ vector to a strictly positive probability vector.

Restriction on totals and the aggregation correction. Expressing the parallel growth assumption in terms of totals and shares:

$$\log n^0_{1,1} - \log n^0_{1,0} = \underbrace{(\log n^0_{0,1} - \log n^0_{0,0})}_{\text{control total growth}} + \underbrace{\log\Big(\sum_{k=1}^{p} \frac{\pi^0_{k,0,1}}{\pi^0_{k,0,0}}\, \pi^0_{k,1,0}\Big)}_{\text{composition adjustment } \log\lambda}, \tag{13}$$

where

$$\lambda = \sum_{k=1}^{p} \frac{\pi^0_{k,0,1}}{\pi^0_{k,0,0}}\, \pi^0_{k,1,0} \tag{14}$$

is a weighted average of the control group's category-level growth rates, with the treated group's pre-treatment shares as weights. To see why the composition adjustment factor matters, consider evaluating an early voting policy. Suppose the control state shows no change in total turnout across elections, but its vote composition shifts: Democratic vote share rises while Republican share falls. Now assume the treated state, before the policy, has an electorate that is disproportionately Republican, the group losing relative share in the control state. Even with flat aggregate turnout in the control state, the treated state's counterfactual turnout would have declined absent the policy, because its composition is weighted toward the shrinking group. In CoDiD, this appears as a composition adjustment factor. Standard DiD, which imposes parallel trends in total turnout, would wrongly assume no counterfactual change and thus underestimate the policy's effect.

Remark 3 (Aggregation correction and bias of log-total DiD)**.** *The factor λ is the source of the aggregation bias in log-total DiD. Standard DiD applied to $\log n_{g,t}$ sets $\lambda = 1$ implicitly, which holds if and only if the treated and control groups share identical pre-treatment compositions ($\pi^0_{1,0} = \pi^0_{0,0}$). Whenever the two groups differ in pre-treatment composition which is almost always the case in practice, log-total DiD is biased, and the bias is directly equal to $\log\lambda$. CoDiD corrects for this bias by working at the category level. The adjustment factor λ can be computed from pre-treatment data. If λ is close to one in a given application, log-total DiD is approximately valid.*

1.5. **Economic Interpretation: Random Utility Model.** I provide a structural economic foundation for parallel growth. Consider a population of decision-makers who each choose one alternative from $\{1, \ldots, p\}$. In the absence of treatment, the utility that a

decision-maker in group g at time t assigns to alternative k is:

$$U^0_{k,g,t} = \mu^0_{k,g,t} + \log n^0_{g,t} - \log\Big(\sum_j e^{\mu^0_{j,g,t}}\Big) + \varepsilon^0_{k,g,t}, \tag{15}$$

where $\mu^0_{k,g,t}$ is the mean utility (relative attractiveness) of alternative k, the second and third terms together constitute the *absolute attractiveness* component (which scales with the group size), and $\varepsilon^0_{k,g,t}$ is an idiosyncratic shock with i.i.d. standard Gumbel errors. The first term reflects how desirable option k is relative to all other alternatives at the same (g, t). The second captures the overall size or participation level (e.g., how many people are making a choice at all). It scales all utilities equally, affecting the total mass of choices but not their relative shares. When it rises (say, due to higher purchasing power, stronger participation, or favorable aggregate conditions), all utilities shift up equally, people are more likely to choose something, but the proportions among categories might remain the same unless the first component also changes. Together, they form a utility concept that blends relative preferences and absolute participation.

Proposition 2 (Parallel growth ⇔ parallel trends in utilities)**.** *Under the RUM in* (15) *with i.i.d. Gumbel errors, and under Assumption 1, Assumption 3 is equivalent to:*

$$\mathbb{E}[U^0_{k,1,1}] - \mathbb{E}[U^0_{k,1,0}] = \mathbb{E}[U^0_{k,0,1}] - \mathbb{E}[U^0_{k,0,0}] \quad \textit{for all } k. \tag{16}$$

This equivalence provides a clean structural interpretation: CoDiD's identifying assumption is that, absent treatment, both groups would have experienced the same evolution in the average attractiveness of each option. The treatment effect then captures the policy's impact on the underlying desirability of each category.

Proposition 3 (Parallel growth ⇔ parallel trends in utility differences)**.** *Under the same conditions as Proposition 2, Assumption 3 implies that: for every pair of categories* (k, j)*,*

$$\big(\mathbb{E}[U^0_{k,1,1}] - \mathbb{E}[U^0_{j,1,1}]\big) - \big(\mathbb{E}[U^0_{k,1,0}] - \mathbb{E}[U^0_{j,1,0}]\big) = \big(\mathbb{E}[U^0_{k,0,1}] - \mathbb{E}[U^0_{j,0,1}]\big) - \big(\mathbb{E}[U^0_{k,0,0}] - \mathbb{E}[U^0_{j,0,0}]\big). \tag{17}$$

Proposition 3 connects to Proposition 1: log-odds are differences of log-quantities, hence differences of expected utilities. Parallel growth in log-quantities implies that pairwise utility differences evolve in parallel absent treatment, the relative preference for one option over another changes at the same rate in both groups.

Remark 4 (CoDiD does not require IIA)**.** *The multinomial logit RUM imposes independence of irrelevant alternatives (IIA). Crucially, IIA is* not required *for the identification result in Theorem 1, which holds for any data-generating process satisfying Assumptions 2–3, regardless of the error structure. The RUM interpretation provides economic motivation for the identifying assumption, but CoDiD's validity is entirely model-agnostic. Appendix C extends the RUM interpretation to Generalized Extreme Value (GEV) error structures, including nested logit and cross-nested logit, which relax IIA.*

1.6. **Geometric Interpretation: Aitchison Simplex Geometry.** Share vectors lie in the probability simplex $\mathcal{S}^{p-1}$. The natural geometry on this space Aitchison geometry (Aitchison, 1982; Egozcue et al., 2003) equips $\mathcal{S}^{p-1}$ with the structure of a real vector space.

Proposition 4 (Aitchison vector space structure, Aitchison, 2002; Egozcue et al., 2003)**.** *The interior of $\mathcal{S}^{p-1}$ is a $(p-1)$-dimensional real vector space under the perturbation operator $(\oplus)$ and powering operator $(\odot)$ defined by:*

$$\pi^1 \oplus \pi^2 = \frac{1}{\sum_j \pi_j^1 \pi_j^2}\big(\pi_1^1\pi_1^2, \ldots, \pi_p^1\pi_p^2\big), \qquad \alpha \odot \pi^1 = \frac{1}{\sum_j (\pi_j^1)^\alpha}\big((\pi_1^1)^\alpha, \ldots, (\pi_p^1)^\alpha\big).$$

The zero element (neutral element for $\oplus$) is the uniform composition $(1/p, \ldots, 1/p)$. The compositional difference is $\pi^1 \ominus \pi^2 = \pi^1 \oplus ((-1) \odot \pi^2)$.

Proposition 5 (Compositional difference interpretation)**.** *Under Assumptions 1 and 3:*

$$\pi_{1,1}^0 \ominus \pi_{1,0}^0 = \pi_{0,1}^0 \ominus \pi_{0,0}^0. \tag{18}$$

That is, the compositional change in shares for the treated group (absent treatment) equals the compositional change in shares for the control group. The compositional treatment effect satisfies:

$$\mathrm{CTT} = (\pi_{1,1}^1 \ominus \pi_{1,0}^0) \ominus (\pi_{0,1}^0 \ominus \pi_{0,0}^0), \tag{19}$$

which is the Aitchison-space analogue of the classical additive ATT formula.

Remark 5 (Aitchison vs. Euclidean parallelism)**.** *Figure 2 depicts the share trajectories in the simplex. The control group's path from $\pi_{0,0}^0$ to $\pi_{0,1}^0$ (red curve) and the treated group's counterfactual path from $\pi_{1,0}^0$ to $\pi_{1,1}^0$ (blue curve) appear curved in Euclidean space because the simplex is a non-linear space. These paths are, however,* straight lines *(geodesics) in*

the Aitchison metric. Parallel growth is a statement about Aitchison *parallelism, not Euclidean parallelism. Two Euclidean-parallel paths on the simplex would in general escape* $\mathcal{S}^{p-1}$*, making the counterfactual infeasible. Aitchison-parallel paths remain in the simplex by construction.*

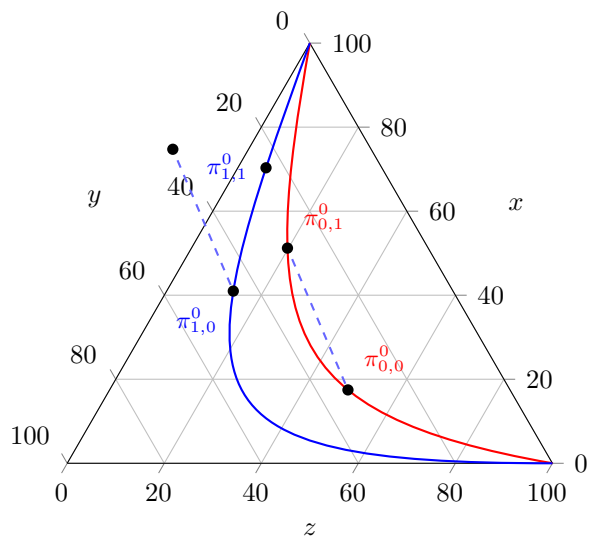

Figure 2. Parallel growth in the simplex (3-category case). The red curve shows the control group's trajectory from $\pi^0_{0,0}$ to $\pi^0_{0,1}$. The blue curve shows the treated group's counterfactual path from $\pi^0_{1,0}$ to $\pi^0_{1,1}$. Dashed lines indicate the parallel translation in Aitchison geometry; these correspond to straight geodesic paths on the simplex.

2. Extensions with Multiple Time Periods

2.1. **Pre-trends Assessment and Testability.** When $T_1 \geq 1$ pre-treatment periods are available, the credibility of Assumption 3 can be assessed. Parallel growth in pre-treatment periods requires that the log-differences $\log q^0_{k,1,t} - \log q^0_{k,0,t}$ be constant over all $t \leq 0$ for each k. A formal Wald test of this restriction is developed in Section 3.3. If the pre-treatment log-quantities appear parallel in a graph, this supports the assumption; if they diverge, it raises concerns. Figure 3 in the application (Section 4) illustrates such a graphical assessment.

2.2. **Partial Identification under Robust Parallel Growth.** When pre-treatment log-differences are not exactly constant, I introduce a relaxation inspired by Ban and Kédagni

(2022). Let the time index run over $t \in \{-T_1, \ldots, -1, 0, 1\}$, with treatment occurring at $t = 1$. Define for each category k the minimum and maximum pre-treatment log-differences:

$$d_k^{\min} = \min_{t \le 0}\big(\log q_{k,1,t}^0 - \log q_{k,0,t}^0\big), \qquad d_k^{\max} = \max_{t \le 0}\big(\log q_{k,1,t}^0 - \log q_{k,0,t}^0\big). \tag{20}$$

Assumption 4 (Robust Parallel Growth)**.** *For each category k, the post-treatment log-difference lies within the range observed in pre-treatment periods:*

$$d_{k,1} \equiv \log q_{k,1,1}^0 - \log q_{k,0,1}^0 \in \big[d_k^{\min},\ d_k^{\max}\big]. \tag{21}$$

When $d_k^{\min} = d_k^{\max}$ for all k, the pre-treatment log-differences are exactly constant and Assumption 3 is recovered as a special case. In general, Assumption 4 is strictly weaker and delivers partial identification.

Theorem 2 (Sharp Partial Identification)**.** *Under Assumptions 1 and 4, define:*

$$b_k^{\min} = \exp\big(d_k^{\min} + \log q_{k,0,1}^0\big), \qquad b_k^{\max} = \exp\big(d_k^{\max} + \log q_{k,0,1}^0\big). \tag{22}$$

The sharp identified set for the counterfactual is:

$$q_{k,1,1}^0 \in \big[b_k^{\min},\ b_k^{\max}\big], \tag{23}$$

$$n_{1,1}^0 \in \Big[\sum_{k=1}^{p} b_k^{\min},\ \sum_{k=1}^{p} b_k^{\max}\Big], \tag{24}$$

$$\pi_{1,1}^0 \in \mathcal{P} = \Big\{\pi \in \mathcal{S}^{p-1} : \pi_k = q_k \big/ \textstyle\sum_j q_j,\ q_k \in [b_k^{\min}, b_k^{\max}] \text{ for all } k\Big\}. \tag{25}$$

‘

Remark 6 (Identified set for ATT)**.** *Since $\pi_{1,1}^1$ is point-identified from the data, the identified set for $\mathrm{ATT}_k = \pi_{k,1,1}^1 - \pi_{k,1,1}^0$ follows directly from (25). Importantly, $\mathcal{P}$ is a* strict subset *of the Cartesian product of the marginal intervals for each $\pi_{k,1,1}^0$, because the simplex constraint $\sum_k \pi_k = 1$ creates cross-category dependence. Thus, computing category-specific bounds independently and taking Cartesian products would over-state the uncertainty.*

Remark 7 (Staggered treatment adoption)**.** *When units adopt treatment at different times, the 2×2 result in Theorem 1 generalizes by applying CoDiD within each cohort-specific 2×2 comparison, using never-treated or not-yet-treated units as controls in the spirit of Callaway and Sant'Anna (2020). Aggregation across cohorts follows the same weighting approach used*

for cohort-specific ATT and GTT parameters. I leave the formal development of staggered adoption for compositional outcomes as a natural avenue for future work.

3. Estimation and Inference

3.1. **Sampling Model.** The primitive objects in Theorem 1 are the log-counts $\log q^0_{k,g,t}$. I therefore build the entire asymptotic theory starting from the *vector of observed counts* for each cell, and derive all other quantities, the counterfactual total, shares, GTT, ATT, and any functional thereof, as smooth functions of those counts.

For each cell $(g,t) \in \{0,1\}^2$, I observe p counts drawn independently from a multinomial distribution:

$$(q_{1,g,t}, \ldots, q_{p,g,t}) \;\sim\; \text{Multinomial}\big(n_{g,t},\; \pi^*_{1,g,t}, \ldots, \pi^*_{p,g,t}\big), \tag{26}$$

where $n_{g,t}$ is the (fixed) cell sample size and $\pi^*_{k,g,t}$ are the true (population) probabilities satisfying $\sum_k \pi^*_{k,g,t} = 1$. Counts across different cells are mutually independent. Let $n = \sum_{g,t} n_{g,t}$ denote the total sample size.

Assumption 5 (Regular Sampling)**.** *As $n \to \infty$: (i) $n_{g,t}/n \to \rho_{g,t} \in (0,1)$ for all (g,t), with $\sum_{g,t} \rho_{g,t} = 1$; and (ii) $\pi^*_{k,g,t} \in (0,1)$ for all k and (g,t), which is the sampling analogue of Assumption 1.*

Condition (i) requires all four cells to grow proportionally with n, ruling out designs in which one cell remains bounded. Condition (ii) ensures that all log-counts are well defined in the population and that no category degenerates. The identification formula expresses each counterfactual log-count as a linear combination of three observed log-counts:

$$\log q^0_{k,1,1} = \log q_{k,1,0} + \log q_{k,0,1} - \log q_{k,0,0}. \tag{27}$$

Substituting the observed counts directly yields, for each category k:

$$\hat{q}^0_{k,1,1} = \exp\big(\log q_{k,1,0} + \log q_{k,0,1} - \log q_{k,0,0}\big). \tag{28}$$

The vector $\hat{q}^0_{1,1} = (\hat{q}^0_{1,1,1}, \ldots, \hat{q}^0_{p,1,1})^\top$ is the *estimated counterfactual count vector*. All other treatment-effect estimators are smooth functions of $\hat{q}^0_{1,1}$ and the directly observed counts in cell $(1,1)$.

Asymptotic distribution of proportions. Let $\hat{\pi}_{g,t} = q_{g,t}/n_{g,t}$ denote the vector of sample proportions in cell (g,t), and let $\pi^*_{g,t}$ denote the true population proportions. By the standard multivariate CLT for multinomial proportions (or directly from the multinomial CLT for counts dividing by $n_{g,t}$), we have

$$\sqrt{n_{g,t}}\left(\hat{\pi}_{g,t} - \pi^*_{g,t}\right) \xrightarrow{d} \mathcal{N}\big(\mathbf{0},\, \Sigma_{g,t}\big), \qquad \Sigma_{g,t} = \operatorname{diag}(\pi^*_{g,t}) - \pi^*_{g,t}(\pi^*_{g,t})^\top. \tag{29}$$

This holds for each cell (g,t) as $n_{g,t} \to \infty$, assuming $\pi^*_{k,g,t} > 0$ for all k. Using Assumption 5, we rescale by a common rate $\sqrt{n}$, write

$$\sqrt{n}\left(\hat{\pi}_{g,t} - \pi^*_{g,t}\right) = \sqrt{\frac{n}{n_{g,t}}} \cdot \sqrt{n_{g,t}}\left(\hat{\pi}_{g,t} - \pi^*_{g,t}\right).$$

Since $\sqrt{n/n_{g,t}} = 1/\sqrt{n_{g,t}/n} \to 1/\sqrt{\rho_{g,t}}$, Slutsky's theorem gives

$$\sqrt{n}\left(\hat{\pi}_{g,t} - \pi^*_{g,t}\right) \xrightarrow{d} \mathcal{N}\Big(\mathbf{0},\, \frac{1}{\rho_{g,t}}\,\Sigma_{g,t}\Big). \tag{30}$$

Thus, the asymptotic variance at the common $\sqrt{n}$ rate is inflated by $1/\rho_{g,t}$ when the cell size is a fraction $\rho_{g,t}$ of the total sample. Since counts across cells are mutually independent, the joint CLT for the vectors $\hat{\pi}_{1,0}, \hat{\pi}_{0,1}, \hat{\pi}_{0,0}$ follows directly from the Cramér–Wold theorem, with independent blocks and the respective covariance matrices $(1/\rho_{g,t})\Sigma_{g,t}$.

Define the componentwise log map $L : \mathbb{R}^p_{++} \to \mathbb{R}^p$ by $L(v)_k = \log v_k$. This map is continuously differentiable at any $v^* \gg \mathbf{0}$ with Jacobian

$$DL(v^*) = \operatorname{diag}\big(1/v^*_1, \ldots, 1/v^*_p\big).$$

Applying the delta method to (30) at the point $\pi^*_{g,t} \gg \mathbf{0}$, we obtain

$$\sqrt{n}\left(\log \hat{\pi}_{g,t} - \log \pi^*_{g,t}\right) \xrightarrow{d} \mathcal{N}\Big(\mathbf{0},\, \frac{1}{\rho_{g,t}}\,A_{g,t}\Big), \tag{31}$$

where

$$A_{g,t} = \operatorname{diag}\Big(\tfrac{1}{\pi^*_{g,t}}\Big)\,\Sigma_{g,t}\,\operatorname{diag}\Big(\tfrac{1}{\pi^*_{g,t}}\Big). \tag{32}$$

The (k,j) entry of $A_{g,t}$ is easily computed as

$$[A_{g,t}]_{kj} = \frac{[\Sigma_{g,t}]_{kj}}{\pi^*_{k,g,t}\,\pi^*_{j,g,t}} = \frac{\mathbf{1}_{\{k=j\}}}{\pi^*_{k,g,t}} - 1. \tag{33}$$

From log-proportions to log-counts. Observe that for each cell (g,t),

$$\log q_{k,g,t} = \log n_{g,t} + \log \hat{\pi}_{k,g,t}, \qquad \log q^*_{k,g,t} = \log n_{g,t} + \log \pi^*_{k,g,t}.$$

Therefore the $\log n_{g,t}$ terms cancel, giving the exact identity

$$\log q_{g,t} - \log q^*_{g,t} = \log \hat{\pi}_{g,t} - \log \pi^*_{g,t}, \tag{34}$$

where the equality is componentwise. Hence the CLT for the centered log-count vector is identical to that for the centered log-proportion vector:

$$\sqrt{n}\left(\log q_{g,t} - \log q^*_{g,t}\right) \xrightarrow{d} \mathcal{N}\Big(\mathbf{0},\ \frac{1}{\rho_{g,t}} A_{g,t}\Big). \tag{35}$$

Combining the three cells. The counterfactual log-count estimator is the linear combination

$$\log \hat{q}^0_{1,1} = \log q_{1,0} + \log q_{0,1} - \log q_{0,0}. \tag{36}$$

By the identification formula (27), the true value satisfies $\log q^{0*}_{1,1} = \log q^*_{1,0} + \log q^*_{0,1} - \log q^*_{0,0}$, so

$$\log \hat{q}^0_{1,1} - \log q^{0*}_{1,1} = \big(\log q_{1,0} - \log q^*_{1,0}\big) + \big(\log q_{0,1} - \log q^*_{0,1}\big) - \big(\log q_{0,0} - \log q^*_{0,0}\big).$$

Since the three cells are independent, the limits from (35) combine additively:

$$\sqrt{n}\left(\log \hat{q}^0_{1,1} - \log q^{0*}_{1,1}\right) \xrightarrow{d} \mathcal{N}\big(\mathbf{0},\ \Sigma_c\big), \tag{37}$$

where

$$\Sigma_c = \frac{A_{1,0}}{\rho_{1,0}} + \frac{A_{0,1}}{\rho_{0,1}} + \frac{A_{0,0}}{\rho_{0,0}}. \tag{38}$$

The three summands reflect the sampling noise from the three observable cells entering the identification formula. The treated cell $(1,1)$ does not appear in Σ_c because it does not enter the counterfactual formula (36).

CLT for the counterfactual count vector. A final application of the delta method, now to the componentwise exponential map $\exp : \mathbb{R}^p \to \mathbb{R}^p_{++}$, whose Jacobian at $\log q^{0*}_{1,1}$ is $\mathrm{diag}(q^{0*}_{1,1})$, yields the CLT for the counterfactual count vector:

$$\sqrt{n}\left(\hat{q}^0_{1,1} - q^{0*}_{1,1}\right) \xrightarrow{d} \mathcal{N}\big(\mathbf{0},\ \Sigma_q\big), \quad \Sigma_q = \mathrm{diag}(q^{0*}_{1,1})\,\Sigma_c\,\mathrm{diag}(q^{0*}_{1,1}). \tag{39}$$

Here the map exp is continuously differentiable, so the standard delta method applies directly.

Remark 8. *The final asymptotic variance Σ_q can be expressed explicitly in terms of the true proportions $\pi^*_{g,t}$ and the limiting fractions $\rho_{g,t}$. Using the entry-wise formula for $A_{g,t}$, we have for any two categories k, j:*

$$[\Sigma_q]_{kj} = q^{0*}_{k,1,1} q^{0*}_{j,1,1} \left(\frac{\mathbf{1}_{\{k=j\}}}{\rho_{1,0}\pi^*_{k,1,0}} + \frac{\mathbf{1}_{\{k=j\}}}{\rho_{0,1}\pi^*_{k,0,1}} + \frac{\mathbf{1}_{\{k=j\}}}{\rho_{0,0}\pi^*_{k,0,0}} - \frac{1}{\rho_{1,0}} - \frac{1}{\rho_{0,1}} - \frac{1}{\rho_{0,0}} \right).$$

This clarifies the contributions of each cell to the variance of the counterfactual count estimator.

Equation (39) is the main result from which all subsequent asymptotic distributions follow from further applications of the delta method.

3.2. **Bootstrap Inference.** The parametric bootstrap is convenient in this setting: it automatically propagates uncertainty through the nonlinear normalization step and produces confidence sets that respect the simplex constraint.

Algorithm. For $b = 1, \ldots, B$, draw independently for each cell (g, t):

$$(q^{(b)}_{1,g,t}, \ldots, q^{(b)}_{p,g,t}) \sim \text{Multinomial}\big(n_{g,t},\ \hat{\pi}_{1,g,t}, \ldots, \hat{\pi}_{p,g,t}\big),$$

For each bootstrap draw, recompute all estimators neededed $\hat{q}^{0,(b)}_{1,1}$, $\hat{n}^{0,(b)}_{1,1}$, $\hat{\pi}^{0,(b)}_{1,1}$, $\hat{\text{GTT}}^{(b)}_k$, $\hat{\text{ATT}}^{(b)}_k$. For any scalar parameter θ, the $(1-\alpha)$ percentile bootstrap confidence interval is given by the empirical $[\alpha/2,\ 1-\alpha/2]$ quantiles of $\{\hat{\theta}^{(b)}\}_{b=1}^B$.

Proposition 6 (Bootstrap Consistency)**.** *Under the conditions of Proposition* **??**, *the parametric bootstrap consistently estimates the asymptotic distribution of all CoDiD estimators. For any scalar parameter $\theta \in \{q^0_{k,1,1},\ n^0_{1,1},\ \pi^0_{k,1,1},\ \text{ATT}_k,\ \text{GTT}_k\}$,*

$$\sup_{x\in\mathbb{R}} \Big| \overset{*}{\Pr}\big(\sqrt{n}(\hat{\theta}^{(b)} - \hat{\theta}) \le x\big) - \Pr\big(\sqrt{n}(\hat{\theta} - \theta^*) \le x\big) \Big| \xrightarrow{p} 0, \tag{40}$$

where $\Pr^*$ denotes probability conditional on the observed data.

3.3. **Pre-trends Test.** With $T_1 \ge 1$ pre-treatment periods, Assumption 3 implies that log-count differences between groups remain constant over time:

$$\Delta^*_{k,t} \equiv \log q^*_{k,1,t} - \log q^*_{k,0,t} = \Delta^*_k, \quad \text{for all } k \text{ and } t \le 0, \tag{41}$$

where $q^*_{k,g,t} = n_{g,t}\, \pi^*_{k,g,t}$. To test this restriction, define the sample log-count differences

$$\hat{\Delta}_{k,t} = \log q_{k,1,t} - \log q_{k,0,t}. \tag{42}$$

Using period $t = 0$ as the reference, construct the centered differences for each pre-treatment period $s \in \{-T_1, \ldots, -1\}$ and category $k = 1, \ldots, p$:

$$\hat{R}_{k,s} = \hat{\Delta}_{k,s} - \hat{\Delta}_{k,0},$$

and stack these into a pT_1-dimensional vector $\hat{R}$. Under Assumption 5, the independence across cells and periods, the log-count CLT derived implies that, under the null hypothesis,

$$\sqrt{n}\,\hat{R} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V}_R),$$

where $\mathbf{V}_R$ is block-diagonal. Each block corresponds to a pre-treatment period and is constructed from the matrices $A_{g,t}$ and the limiting sample share parameters $\rho_{g,t}$ defined earlier. A consistent estimator $\hat{\mathbf{V}}_R$ is obtained by plugging in sample analogs $\hat{\pi}_{g,t}$ and $\hat{\rho}_{g,t} = n_{g,t}/n$.

The Wald statistic

$$W = n\,\hat{R}^\top \hat{\mathbf{V}}_R^{-1} \hat{R} \tag{43}$$

satisfies

$$W \xrightarrow{d} \chi^2_{pT_1} \quad \text{under } H_0,$$

provided $\hat{\mathbf{V}}_R$ is consistent. Large values of W indicate violations of pre-treatment parallel growth, suggesting that Assumption 3 may be violated and motivating the partial- identification approach developed in Section 2.2.

Remark 9 (Event-study analogue and graphical assessment)**.** *For each category k, plotting $\hat{\Delta}_{k,t}$ against time t provides a graphical complement to the formal Wald test. Parallel pre-trends manifest as horizontally flat trajectories across pre-treatment periods, whereas systematic deviations suggest violations of the identifying assumption. This event-study style visualization is particularly informative when $T_1 \geq 2$, as it reveals the pattern and magnitude of any pre-existing differential trends.*

4. Application: Early Voting and the 2008 U.S. Presidential Election

4.1. **Research Design.** Early voting allows registered voters to cast ballots before Election Day, either in person or by mail. I evaluate the causal effect of early-voting programs on both total turnout and the compositional distribution of votes across parties in the 2008 U.S. presidential election.

The treated group consists of Maryland and New Jersey, which introduced early voting before the 2008 election. Pennsylvania and New York serve as the control group. These states are geographically contiguous, share comparable demographic and political profiles (Table 1), and consistently supported Democratic presidential candidates from 1992 to 2004, providing a plausible basis for the parallel growth assumption.

CoDiD requires three conditions. First, *parallel growth* for log-vote counts: Figure 3 documents broadly parallel pre-treatment log-count trajectories from 1992 to 2004 across all three party categories, supporting this assumption. Second, *no interference* (SUTVA): voters in one state are unlikely to change their party choice in response to early-voting laws in a neighboring state. Third, *no anticipation*: early-voting programs were not announced years in advance, so voters had no opportunity to adjust pre-2008 behavior in anticipation of the 2008 reform.

Why standard approaches fail here. Figure 4 shows that raw vote shares do not exhibit parallel trends. Figure 5 shows that the large-scale difference between groups (the control group is roughly twice the size of the treated group) makes raw-quantity parallel trends implausible. CoDiD's parallel growth assumption operates on log-quantities, correcting for both issues simultaneously.

Table 1. Population Composition in 2007

State/Group	**Total Pop.**	**White**	**Black**	**Am.Ind.**	**Asian**	**Two+**	**Hisp./Latino**	**White NH**
Maryland	5,618,344	63.6%	29.5%	0.3%	5.0%	1.6%	6.3%	58.1%
New Jersey	8,685,920	76.3%	14.5%	0.3%	7.5%	1.3%	15.9%	62.2%
MD+NJ	14,304,264	71.3%	20.8%	0.3%	6.5%	1.4%	12.1%	60.5%
New York	19,297,729	73.6%	17.3%	0.5%	6.9%	1.5%	16.4%	60.3%
Pennsylvania	12,432,792	85.6%	10.8%	0.2%	2.4%	1.0%	4.5%	81.8%
NY+PA	31,730,521	78.0%	14.5%	0.4%	5.0%	1.3%	11.7%	69.9%

Source: U.S. Census Bureau, Population Division, May 2008.

4.2. **Results.**

Growth Treatment Effects (GTT).. Table 2 reports estimated GTT with 95% bootstrap confidence intervals ($B = 9{,}999$ draws).

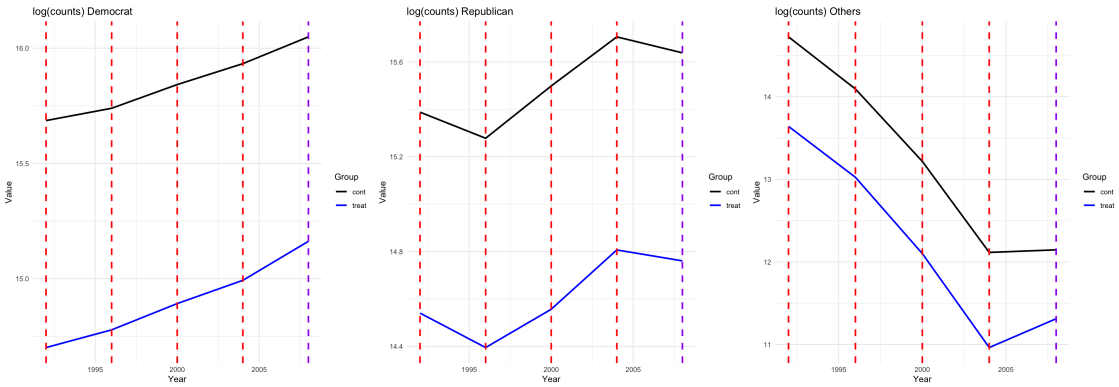

Figure 3. Log-count evolution, 1992–2008 (treated vs. control). The dotted red vertical line separates pre-treatment elections (1992–2004) from the post-treatment election (2008). The broadly parallel trajectories in the pre-treatment period support the plausibility of the parallel growth assumption.

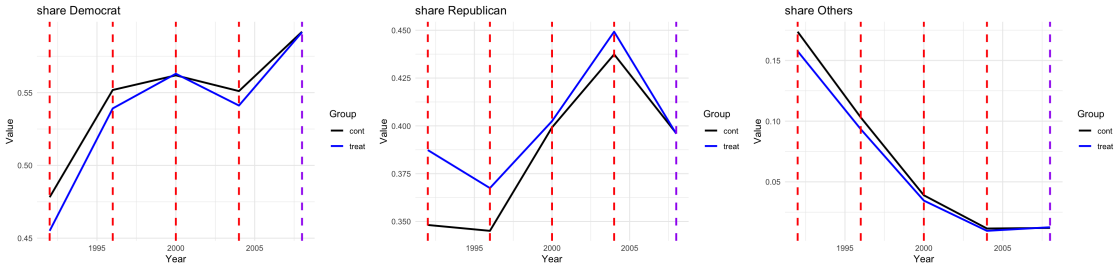

Figure 4. Raw vote-share evolution, 1992–2008 (treated vs. control). The non-parallel pre-treatment share trajectories illustrate why linear parallel trends on shares is inappropriate.

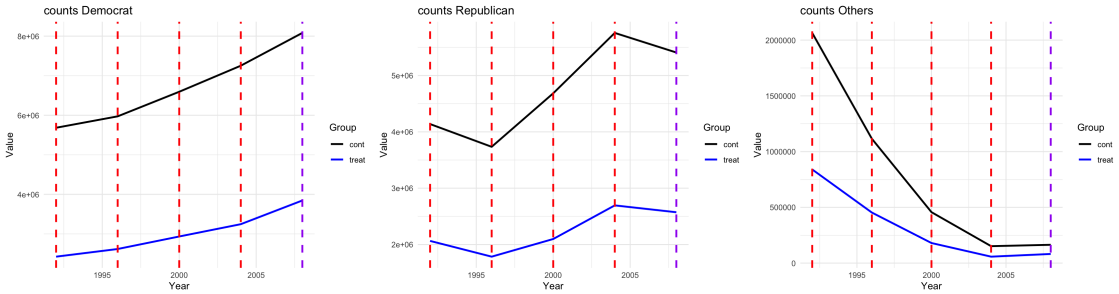

Figure 5. Raw-count evolution, 1992–2008 (treated vs. control). The large scale difference between groups makes additive parallel trends on raw quantities implausible.

The policy increased total turnout by approximately 4.4%, consistent with the existing literature on early voting (Berry et al., 2025). All three categories experienced statistically

Table 2. Estimated Growth Treatment Effects on the Treated (GTT)

Party	**Estimate**	**[95% CI]**
Democrat	0.0545	[0.0533, 0.0557]
Republican	0.0214	[0.0199, 0.0230]
Other	0.3747	[0.3578, 0.3915]
Total	0.0441	[0.0441, 0.0443]

significant increases. Democrats gained 5.5%, Republicans 2.1%, and third-party candidates 37.5% (the latter reflecting a small base). The differential gain for Democrats is consistent with evidence that early voting disproportionately mobilizes left-leaning voters (Berry et al., 2025).

Average Treatment Effects on Shares (ATT).. Table 3 reports the observed distribution, counterfactual distribution, and ATT.

Table 3. Estimated Categorical Distributions and ATT

Party	**Observed [CI]**	**Counterfactual [CI]**	**ATT [CI]**
Democrat	0.5915 [0.5911, 0.5915]	0.5823 [0.5818, 0.5829]	+0.0092 [0.0086, 0.0099]
Republican	0.3959 [0.3954, 0.3962]	0.4076 [0.4071, 0.4082]	−0.0117 [−0.0124, −0.0111]
Other	0.0126 [0.0125, 0.0127]	0.0101 [0.0100, 0.0102]	+0.0025 [0.0024, 0.0027]

Early voting increased the Democratic share by 0.92 pp (from a counterfactual of 58.23% to an observed 59.15%), decreased the Republican share by 1.17 pp (from 40.76% to 39.59%), and increased the third-party share by 0.25 pp. All effects are precisely estimated. By construction, the ATT effects sum to zero across parties, confirming internal consistency. Note that these results are not identifiable by any collection of separate binary DiDs: the counterfactual shares in Table 3 are the *unique* feasible vector consistent with parallel growth.

Compositional Treatment Effects (CTT).. Table 4 reports the CTT, which measures the *relative* redistribution of probability mass.

Table 4. Estimated Compositional Treatment Effect on the Treated (CTT)

Party	**Estimate**	**[95% CI]**
Democrat	0.3136	[0.3119, 0.3151]
Republican	0.2998	[0.2982, 0.3012]
Other	0.3867	[0.3838, 0.3898]

Each entry gives the probability of supporting a given party under a hypothetical equal baseline of 1/3 each. Third-party support rises most relative to the baseline (38.7% vs. 33.3%), while Republicans lose the most relative importance (30.0%). While both major parties lose relative ground to third parties, the absolute ranking of the parties (Democrats first, Republicans second, Others third) is unchanged by treatment.

Sensitivity and partial identification bounds. I apply Theorem 2 using all pre-treatment elections (1992–2004) to construct the bounds $[d_k^{\min}, d_k^{\max}]$. The resulting bounds on the ATT are: Democrats $[0.007, 0.012]$ pp, Republicans $[-0.014, -0.009]$ pp. The point estimates from Table 3 lie comfortably within the identified sets, confirming that the estimates are robust to deviations from exact parallel growth within the range of historical pre-treatment variation.

5. Conclusion

I introduce Compositional Difference-in-Differences (CoDiD), a framework for causal inference on vectors of category-specific quantities in DiD settings. Starting from the observation that standard practice running separate linear DiD regressions for each category or for one selected category is simultaneously misspecified, incoherent across categories, and incapable of recovering the total, I develop a unified alternative that solves all three problems at once. The key insight is to work with log-quantities rather than raw quantities or proportions. The parallel growth assumption, parallel trends in log-quantities, is the natural multiplicative counterpart of additive parallel trends. It delivers point identification of the full counterfactual compositional vector through a simple closed-form formula, from which both intensive-margin effects (ATT, CTT) and extensive-margin effects (GTT) are immediately recovered. The assumption has a precise structural interpretation as parallel

trends in latent expected utilities under a standard random-utility model, and a geometric interpretation as parallel paths in the Aitchison metric on the probability simplex.

The estimator is $\sqrt{n}$-consistent and asymptotically normal under multinomial sampling; I derive exact asymptotic variance formulas and prove bootstrap consistency. A formal Wald pre-trends test and a partial-identification relaxation make the approach robust in practice.

Several natural extensions remain. First, incorporating continuous covariates would require nonparametric or semiparametric smoothing and is left for future work. Second, a formal treatment of staggered adoption in the compositional setting, analogous to Callaway and Sant'Anna (2020), would extend the framework to modern panel designs.

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Appendix

Appendix A. Proofs

A.1. **Proof of Theorem 1.** Starting from Assumption 3: for each k, $\log q^0_{k,1,1} - \log q^0_{k,1,0} = \log q^0_{k,0,1} - \log q^0_{k,0,0}$. Adding $\log q^0_{k,1,0}$ to both sides and exponentiating (valid by Assumption 1): $q^0_{k,1,1} = q_{k,1,0} \cdot q_{k,0,1}/q_{k,0,0}$. Summing over k gives $n^0_{1,1}$, and normalizing gives $\pi^0_{k,1,1} = q^0_{k,1,1}/n^0_{1,1}$. □

A.2. **Proof of proposition 1.** We have:

$$U^0_{k,g,t} = \mu^0_{k,g,t} - \log\left(\sum_k e^{\mu^0_{k,g,t}}\right) + \log\left(n^0_{g,t}\right) + \varepsilon^0_{k,g,t}$$

In the standard multinomial logit RUM:

$$\pi_{k,g,t} = \Pr\left(U^0_{k,g,t} \geq U^0_{j,g,t} \text{ for all } j \neq k\right).$$

Since $\varepsilon^0_{k,g,t} \sim$ i.i.d. Gumbel, the choice probability has the closed-form logit formula:

$$\pi_{k,g,t} = \frac{\exp\left(\mu^0_{k,g,t} + \log(n^0_{g,t}) - \log(\sum_j e^{\mu^0_{j,g,t}})\right)}{\sum_j \exp\left(\mu^0_{j,g,t} + \log(n^0_{g,t}) - \log(\sum_l e^{\mu^0_{l,g,t}})\right)}.$$

This simplifies to:

$$\pi_{k,g,t} = \frac{e^{\mu^0_{k,g,t}} \cdot \frac{n^0_{g,t}}{\sum_l e^{\mu^0_{l,g,t}}}}{\sum_j e^{\mu^0_{j,g,t}} \cdot \frac{n^0_{g,t}}{\sum_l e^{\mu^0_{l,g,t}}}} = \frac{e^{\mu^0_{k,g,t}}}{\sum_j e^{\mu^0_{j,g,t}}}.$$

The $n^0_{g,t}$ term cancels out in the probability, which shows that the relative shares depend only on the relative utilities, while $n^0_{g,t}$ scales the absolute number of choices. The number of individuals choosing c_k is:

$$q^0_{k,g,t} = \pi_{k,g,t} \cdot n^0_{g,t} = \frac{e^{\mu^0_{k,g,t}}}{\sum_j e^{\mu^0_{j,g,t}}} \cdot n^0_{g,t}.$$

This leads to:

$$\log q^0_{k,g,t} = \mu^0_{k,g,t} - \log\left(\sum_{k=1}^{p} e^{\mu^0_{k,g,t}}\right) + \log(n^0_{g,t}).$$

Now, we have :

$$\begin{aligned}\mathbb{E}[U^0_{k,g,1}] - \mathbb{E}[U^0_{k,g,0}] &= \mu^0_{k,g,1} - \log(\sum_k e^{\mu^0_{k,g,1}}) + \log\left(n^0_{g,1}\right) + \gamma \\ &\quad -\mu^0_{k,g,0} - \log(\sum_k e^{\mu^0_{k,g,0}}) + \log\left(n^0_{g,0}\right) - \gamma \\ &= \log q^0_{k,g,1} - \log q^0_{k,g,0}\end{aligned}$$

From there, we have that assumption 3 is equivalent to parallel trends in expected utilities:

$$\mathbb{E}[U^0_{k,1,1}] - \mathbb{E}[U^0_{k,1,0}] = \mathbb{E}[U^0_{k,0,1}] - \mathbb{E}[U^0_{k,0,0}], \quad \forall k$$

A.3. **Proof of Proposition 2.** Pick a baseline category cp. The log-odds for each category can be written as :

$$\log\left(\frac{\pi^0_{k,g,t}}{\pi^0_{p,g,t}}\right) = \log\left(\frac{q^0_{k,g,t}}{n^0_{g,t}} \times \frac{n^0_{g,t}}{q^0_{p,g,t}}\right) = \log(q^0_{k,g,t}) - \log(q^0_{p,g,t}).$$

Parallel growths implies hold for each category, so we have :

$$\log(q^0_{k,1,1}) - \log(q^0_{k,1,0}) = \log(q^0_{k,0,1}) - \log(q^0_{k,0,0}),$$

and also

$$\log(q^0_{p,1,1}) - \log(q^0_{p,1,0}) = \log(q^0_{p,0,1}) - \log(q^0_{p,0,0}).$$

Taking the difference between the two equations gives:

$$\begin{gathered}\left\{\log(q^0_{k,1,1}) - \log(q^0_{p,1,1})\right\} - \left\{\log(q^0_{k,1,0}) - \log(q^0_{p,1,0})\right\} = \\ \left\{\log(q^0_{k,0,1}) - \log(q^0_{p,0,1})\right\} - \left\{\log(q^0_{k,0,0}) - \log(q^0_{p,0,0})\right\}\end{gathered}$$

This is exactly parallel trends in log-odds :

$$\log\left(\frac{\pi^0_{k,1,1}}{\pi^0_{l,1,1}}\right) - \log\left(\frac{\pi^0_{k,0,1}}{\pi^0_{l,0,1}}\right) = \log\left(\frac{\pi^0_{k,1,0}}{\pi^0_{l,1,0}}\right) - \log\left(\frac{\pi^0_{k,0,0}}{\pi^0_{l,0,0}}\right).$$

From part 1, I can also derive that:

$$\begin{gathered}\left\{\log(q^0_{k,1,1}) - \log(q^0_{p,1,1})\right\} - \left\{\log(q^0_{k,1,0}) - \log(q^0_{p,1,0})\right\} = \\ \left\{\log(q^0_{k,0,1}) - \log(q^0_{p,0,1})\right\} - \left\{\log(q^0_{k,0,0}) - \log(q^0_{p,0,0})\right\}\end{gathered}$$

is equivalent to

$$\Big\{\mathbb{E}[U^0_{k,1,1}] - \mathbb{E}[U^0_{11}]\Big\} - \Big\{\mathbb{E}[U^0_{k,1,0}] - \mathbb{E}[U^0_{k,1,0}]\Big\} = \\ \Big\{\mathbb{E}[U^0_{k,0,1}] - \mathbb{E}[U^0_{k,0,1}]\Big\} - \Big\{\mathbb{E}[U^0_{k,0,0}] - \mathbb{E}[U^0_{l,0,0}]\Big\}, \tag{44}$$

A.4. **Proof of Proposition 5.** For each k: $\pi^0_{k,1,1}/\pi^0_{k,1,0} = (q^0_{k,1,1}/n^0_{1,1})/(q^0_{k,1,0}/n^0_{1,0}) = (q^0_{k,1,1}/q^0_{k,1,0}) \cdot (n^0_{1,0}/n^0_{1,1})$. By parallel growth, $q^0_{k,1,1}/q^0_{k,1,0} = q^0_{k,0,1}/q^0_{k,0,0}$. The factor $n^0_{1,0}/n^0_{1,1}$ is common across all k and cancels upon normalization in the $\ominus$ operator, yielding $\pi^0_{1,1} \ominus \pi^0_{1,0} = \pi^0_{0,1} \ominus \pi^0_{0,0}$. □

A.5. **Proof of Theorem 2.** We proceed in four steps.

Step 1: Apply Assumption 4. By the Robust Parallel Growth assumption (Assumption 4), for each category $k = 1, \ldots, p$,

$$d_{k,1} = \log\big(q^0_{k,1,1}\big) - \log\big(q^0_{k,0,1}\big) \in \big[\, d_k^{\min},\ d_k^{\max} \,\big].$$

Step 2: Bound the counterfactual quantity $q^0_{k,1,1}$. Exponentiating the inequality yields

$$\exp\big(d_k^{\min}\big) \le \frac{q^0_{k,1,1}}{q^0_{k,0,1}} \le \exp(d_k^{\max})\,.$$

Multiplying through by $q^0_{k,0,1} > 0$ gives

$$\exp\big(d_k^{\min} + \log q^0_{k,0,1}\big) \le q^0_{k,1,1} \le \exp\big(d_k^{\max} + \log q^0_{k,0,1}\big)\,.$$

Define

$$b_k^{\min} = \exp\big(d_k^{\min} + \log q^0_{k,0,1}\big)\,, \quad b_k^{\max} = \exp\big(d_k^{\max} + \log q^0_{k,0,1}\big)\,.$$

Then

$$q^0_{k,1,1} \in \big[\, b_k^{\min},\ b_k^{\max} \,\big],$$

which establishes the first claim.

Step 3: Bound the total mass $n^0_{1,1}$. The total counterfactual mass is $n^0_{1,1} = \sum_{k=1}^p q^0_{k,1,1}$. Since each $q^0_{k,1,1}$ is bounded independently, the sharp bounds on the sum are obtained by summing the individual extrema:

$$\sum_{k=1}^p b_k^{\min} \le n^0_{1,1} \le \sum_{k=1}^p b_k^{\max}.$$

Step 4: Characterize the counterfactual distribution $\pi^0_{1,1}$. The counterfactual probability vector is defined by

$$\pi^0_{1,1,k} = \frac{q^0_{k,1,1}}{n^0_{1,1}}, \quad k = 1, \ldots, p.$$

Because each $q^0_{k,1,1} \in [b^{\min}_k, b^{\max}_k]$ and $n^0_{1,1} = \sum_{j=1}^p q^0_{j,1,1}$, the set of feasible distributions is precisely

$$\left\{ \pi \in \mathcal{S}^{p-1} : \pi_k = \frac{q_k}{\sum_{j=1}^p q_j},\ q_k \in [b^{\min}_k, b^{\max}_k] \text{ for all } k \right\},$$

where $\mathcal{S}^{p-1}$ denotes the $(p-1)$-dimensional simplex.

A.6. **Proof of proposition 6.** For each cell (g,t), the parametric bootstrap draws $\hat{\pi}^{(b)}_{g,t} = q^{(b)}_{g,t}/n_{g,t}$ from Multinomial$(n_{g,t}, \hat{\pi}_{g,t})/n_{g,t}$, where $\hat{\pi}_{g,t}$ is the observed proportion vector. By the classical bootstrap theorem for multinomial data (Bickel & Freedman, 1981), conditional on the data and for almost every sequence of observed proportions converging to $\pi^*_{g,t}$, we have

$$\sqrt{n_{g,t}}\big(\hat{\pi}^{(b)}_{g,t} - \hat{\pi}_{g,t}\big) \xrightarrow{d} \mathcal{N}\big(\mathbf{0},\, \Sigma_{g,t}\big) \quad \text{in probability,} \tag{45}$$

where $\Sigma_{g,t} = \operatorname{diag}(\pi^*_{g,t}) - \pi^*_{g,t}(\pi^*_{g,t})^\top$. Convergence in probability of the conditional distribution means that for any continuous bounded function f,

$$\mathbb{E}^*[f(\sqrt{n_{g,t}}(\hat{\pi}^{(b)}_{g,t} - \hat{\pi}_{g,t}))] \xrightarrow{p} \mathbb{E}[f(Z)],$$

with $Z \sim \mathcal{N}(0, \Sigma_{g,t})$.

Since $n_{g,t}/n \to \rho_{g,t} \in (0,1)$, we have

$$\sqrt{n}(\hat{\pi}^{(b)}_{g,t} - \hat{\pi}_{g,t}) = \sqrt{\frac{n}{n_{g,t}}}\ \sqrt{n_{g,t}}(\hat{\pi}^{(b)}_{g,t} - \hat{\pi}_{g,t}) \xrightarrow{d} \mathcal{N}\left(\mathbf{0},\, \frac{1}{\rho_{g,t}}\Sigma_{g,t}\right) \quad \text{in probability,}$$

by Slutsky's theorem applied conditionally, because $\sqrt{n/n_{g,t}} \to 1/\sqrt{\rho_{g,t}}$ almost surely.

Each CoDiD estimator is a smooth function of the cell-specific proportion vectors. Specifically, define the stacked vector

$$\hat{\pi} = (\hat{\pi}^\top_{1,0}, \hat{\pi}^\top_{0,1}, \hat{\pi}^\top_{0,0})^\top \in \mathbb{R}^{3p},$$

with true value $\pi^* = ((\pi^*_{1,0})^\top, (\pi^*_{0,1})^\top, (\pi^*_{0,0})^\top)^\top$. Each parameter of interest can be written as $\theta = h(\pi^*)$ for a continuously differentiable function h in a neighborhood of π^*, and its estimator is $\hat{\theta} = h(\hat{\pi})$.

By the parametric delta method for the bootstrap (Theorem 1.3.9 in van der Vaart & Wellner, 1996), if h is continuously differentiable at π^* and $\sqrt{n}(\hat{\pi}^{(b)} - \hat{\pi}) \rightsquigarrow \mathcal{N}(0, \Sigma_\pi)$ in probability, then

$$\sqrt{n}(h(\hat{\pi}^{(b)}) - h(\hat{\pi})) \rightsquigarrow \mathcal{N}(0, Dh(\pi^*)\Sigma_\pi Dh(\pi^*)^\top) \quad \text{in probability,}$$

where $\rightsquigarrow$ denotes weak convergence conditional on the data. The asymptotic covariance matrix matches that from the original delta method applied to $\sqrt{n}(\hat{\pi} - \pi^*)$.

For scalar parameters, the conditional distribution of $\sqrt{n}(\hat{\theta}^{(b)} - \hat{\theta})$ converges in probability to $\mathcal{N}(0, \sigma_\theta^2)$, where σ_θ^2 is the asymptotic variance of $\sqrt{n}(\hat{\theta} - \theta^*)$.

The convergence in distribution conditional on the data implies the stronger statement of uniform convergence of distribution functions (Polya's theorem) because the limiting distribution is continuous (normal with positive variance). For almost every sequence of data,

$$\sup_{x \in \mathbb{R}} \left| \overset{*}{\Pr}\big(\sqrt{n}(\hat{\theta}^{(b)} - \hat{\theta}) \leq x\big) - \Phi(x/\sigma_\theta) \right| \to 0,$$

and the same holds for the original sampling distribution:

$$\sup_{x \in \mathbb{R}} \left| \Pr\big(\sqrt{n}(\hat{\theta} - \theta^*) \leq x\big) - \Phi(x/\sigma_\theta) \right| \to 0.$$

The triangle inequality then yields

$$\sup_{x \in \mathbb{R}} \left| \overset{*}{\Pr}\big(\sqrt{n}(\hat{\theta}^{(b)} - \hat{\theta}) \leq x\big) - \Pr\big(\sqrt{n}(\hat{\theta} - \theta^*) \leq x\big) \right| \xrightarrow{p} 0.$$

The convergence in probability holds because the exceptional set where the conditional distributions do not converge as the probability tends to zero.

Since each parameter is a smooth function of the same underlying proportion vectors, and the number of parameters is finite, the consistency holds jointly. The continuous mapping theorem applied to the joint conditional distribution ensures that any finite collection of scalar parameters satisfies the uniform convergence property. The covariance structure is preserved by the bootstrap, so percentile confidence intervals inherit correct asymptotic coverage.

Appendix B. Covariate Adjustment

Let $X \in \mathcal{X}$ be a vector of discrete covariates. Stratifying on $X = x$ yields the conditional parallel growth assumption.

Assumption 6 (Conditional Parallel Growth)**.** *For each covariate value $x \in \mathcal{X}$:*

$$\log q^0_{k,1,1,x} - \log q^0_{k,1,0,x} = \log q^0_{k,0,1,x} - \log q^0_{k,0,0,x} \quad \textit{for all } k.$$

The conditional counterfactual is:

$$q^0_{k,1,1,x} = \frac{q_{k,1,0,x} \cdot q_{k,0,1,x}}{q_{k,0,0,x}},$$

and the unconditional counterfactual aggregates over the covariate distribution:

$$q^0_{k,1,1} = \sum_{x \in \mathcal{X}} \mathbb{P}(X = x)\, q^0_{k,1,1,x}, \qquad \pi^0_{k,1,1} = \sum_{x \in \mathcal{X}} \mathbb{P}(X = x)\, \pi^0_{k,1,1,x}.$$

Conditional parallel growth is a weaker assumption than unconditional parallel growth: it allows the counterfactual growth rate to vary with X, requiring only that the treated and control groups share the same conditional growth rate for each value of X. The extension to continuous covariates requires nonparametric or semiparametric smoothing and is left for future work.

Appendix C. Extension to General Random Utility Models

I relax the independence of irrelevant alternatives (IIA) assumption by extending the RUM to Generalized Extreme Value (GEV) errors (McFadden, 1977). Assume the joint distribution of $(\varepsilon^0_1, \ldots, \varepsilon^0_p)$ has the GEV form:

$$F(\varepsilon^0_1, \ldots, \varepsilon^0_p) = \exp\Big[-G(e^{-\varepsilon^0_1}, \ldots, e^{-\varepsilon^0_p})\Big],$$

where $G\colon \mathbb{R}^p_+ \to \mathbb{R}_+$ is a homogeneous generating function. When $G(\mathbf{y}) = \sum_k y_k$ this collapses to the multinomial logit. More general G (nested logit, cross-nested logit, paired combinatorial logit) allow arbitrary correlation across alternatives, relaxing IIA. Under the GEV model, choice probabilities are:

$$\pi^0_{k,g,t} = \frac{\exp(V^0_{k,g,t})\, G_k(\mathbf{V}^0_{g,t})}{G(\mathbf{V}^0_{g,t})}, \quad G_k = \partial G / \partial y_k.$$

Assumption 7 (Parallel growth in GEV adjustment)**.** *For each k:*

$$\log G_k(\mu^0_{1,1}) - \log G_k(\mu^0_{1,0}) = \log G_k(\mu^0_{0,1}) - \log G_k(\mu^0_{0,0}).$$

Under Assumption 7, Proposition 2 extends to the GEV setting: Assumption 3 is equivalent to parallel trends in expected utilities under the GEV model. In the multinomial logit case, $G_k(\mathbf{V}) \equiv 1$, so Assumption 7 is automatically satisfied. More generally, it holds approximately when the correlation structure of preferences does not change over time a natural condition in many panel DiD applications. Table 5 summarizes common GEV models.

Table 5. Examples of GEV Models and Their Generating Functions

Model	$G(\mathbf{y})$	$\pi^0_{k,g,t}$
Logit	$\sum_{j=1}^{p} e^{V_j}$	$\frac{e^{V_k}}{\sum_j e^{V_j}}$
Nested Logit	$\sum_{m=1}^{M}\Big(\sum_{j\in\mathcal{B}_m} e^{V_j/\lambda_m}\Big)^{\lambda_m}$	$\frac{e^{V_k/\lambda_m}\big(\sum_{j\in\mathcal{B}_m} e^{V_j/\lambda_m}\big)^{\lambda_m-1}}{\sum_r\big(\sum_{j\in\mathcal{B}_r} e^{V_j/\lambda_r}\big)^{\lambda_r}}$
PCL	$\sum_{r<s}\big(e^{V_r/\lambda_{rs}} + e^{V_s/\lambda_{rs}}\big)^{\lambda_{rs}}$	$\frac{\sum_{j\neq k} e^{V_k/\lambda_{kj}}\big(e^{V_k/\lambda_{kj}} + e^{V_j/\lambda_{kj}}\big)^{\lambda_{kj}-1}}{\sum_{r<s}\big(e^{V_r/\lambda_{rs}} + e^{V_s/\lambda_{rs}}\big)^{\lambda_{rs}}}$
GNL	$\sum_{m=1}^{M}\Big(\sum_{j\in\mathcal{B}_m} (\alpha_{jm}e^{V_j})^{1/\lambda_m}\Big)^{\lambda_m}$	$\frac{(\alpha_{km}e^{V_k})^{1/\lambda_m}\big(\sum_{j\in\mathcal{B}_m}(\alpha_{jm}e^{V_j})^{1/\lambda_m}\big)^{\lambda_m-1}}{\sum_r\big(\sum_{j\in\mathcal{B}_r}(\alpha_{jr}e^{V_j})^{1/\lambda_r}\big)^{\lambda_r}}$